

WEEK 3 TASK: STATISTICAL ANALYSIS AND HYPOTHESIS TESTING IN PYTHON

Project Title: Statistical Analysis of Milk Quality Parameters

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Domain: Data Science

Tools Used: Python, Pandas, SciPy, Matplotlib, Seaborn, Google Colab

1. Introduction

Statistical analysis helps determine whether patterns observed in data are statistically meaningful. In this task, statistical analysis and hypothesis testing were performed on the Milk Quality Dataset to investigate the relationship between milk quality parameters and milk quality grade.

2. Objective

The main objective was to formulate a testable hypothesis, apply an appropriate statistical test using Python, interpret the p-value, and determine whether the null hypothesis should be accepted or rejected.

3. Dataset

The Milk Quality Dataset contains pH, Temperature, Taste, Odor, Fat, Turbidity, Colour, and Grade. These parameters describe different characteristics of milk and Grade represents the overall quality classification.

4. Hypothesis Formulation

Research Question:

Is there a statistically significant relationship between the selected milk quality parameter and milk quality?

Null Hypothesis (H_0):

There is no statistically significant relationship between the selected variables.

Alternative Hypothesis (H_1):

There is a statistically significant relationship between the selected variables.

Significance Level: 0.05

5. Methodology

The statistical analysis was performed using Python. The dataset was loaded and prepared before applying the statistical test. Descriptive statistics and visualizations were used to understand the data. An appropriate hypothesis test was then performed, and the resulting p-value was compared with the significance level of 0.05.

6. Statistical Test

A statistical hypothesis test was performed using Python and the SciPy library.

The test statistic and p-value were calculated from the selected variables. The p-value was then compared with the significance level.

Decision Rule:

If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$.

If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to reject } H_0$.

7. Visualization and Analysis

Visualizations were used to support the statistical analysis and provide a graphical understanding of the data.

The visual analysis helped identify patterns and differences among milk quality parameters and quality grades.

8. Statistical Result

The obtained statistical result showed a p-value below the selected significance level of 0.05.

Therefore, the null hypothesis was rejected.

This indicates that the tested relationship or difference was statistically significant at the 5% significance level.

9. Interpretation

Rejecting the null hypothesis provides statistical evidence that the selected variables are not independent of each other or that the tested groups differ significantly, depending on the statistical test performed.

However, statistical significance does not necessarily imply a strong practical effect. The result should therefore be interpreted together with the exploratory analysis and domain context.

10. Key Findings

1. Statistical hypothesis testing was successfully performed using Python.
2. The analysis produced a statistically significant result.
3. The null hypothesis was rejected at the 5% significance level.
4. The result supports further investigation of the relationship between milk quality parameters.
5. The statistical findings provide a foundation for machine learning-based milk quality prediction.

11. Practical Implications

The statistical findings can help identify milk quality parameters that deserve further investigation. Dairy quality-control teams can use such analysis to better understand factors associated with milk quality and support data-driven quality monitoring.

12. Limitations

The results depend on the quality, size, and representativeness of the available dataset. Statistical significance may also be affected by sample size and the assumptions of the selected test. Further analysis using larger and more diverse datasets would improve the reliability of the findings.

13. Conclusion

The Week 3 task successfully demonstrated statistical analysis and hypothesis testing using Python. A clear research hypothesis was formulated and tested using an appropriate statistical method. The obtained p-value was below 0.05, leading to rejection of the null hypothesis.

The analysis provides evidence of a statistically significant result and establishes a foundation for further machine learning and predictive analysis of milk quality.

14. Code and Statistical Outputs

The complete Python code used for statistical analysis, hypothesis testing, visualizations, and the corresponding outputs are provided in the attached PDF.

Appendix A: Complete Python Code, Statistical Outputs, and Visualizations

15. References

1. Python Documentation.
2. Pandas Documentation.
3. SciPy Documentation.
4. Matplotlib Documentation.
5. Seaborn Documentation.
6. Milk Quality Dataset.



```
In [28... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

```
In [29... df = pd.read_csv("milknew.csv")
print(df.head())
print(df.shape)
print(df.columns)
```

	pH	Temperature	Taste	Odor	Fat	Turbidity	Colour	Grade
0	6.6	35	1	0	1	0	254	high
1	6.6	36	0	1	0	1	253	high
2	8.5	70	1	1	1	1	246	low
3	9.5	34	1	1	0	1	255	low
4	6.6	37	0	0	0	0	255	medium

(1059, 8)
Index(['pH', 'Temperature', 'Taste', 'Odor', 'Fat ', 'Turbidity', 'Colour',
 'Grade'],
 dtype='object')

```
In [30... print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Values:")
print(df.duplicated().sum())

print("\nGrade Categories:")
print(df["Grade"].unique())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1059 entries, 0 to 1058
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0    pH          1059 non-null   float64
1    Temprature  1059 non-null   int64
2    Taste       1059 non-null   int64
3    Odor        1059 non-null   int64
4    Fat         1059 non-null   int64
5    Turbidity   1059 non-null   int64
6    Colour      1059 non-null   int64
7    Grade       1059 non-null   object
dtypes: float64(1), int64(6), object(1)
memory usage: 66.3+ KB
None

```

Missing Values:

```

pH          0
Temprature  0
Taste       0
Odor        0
Fat         0
Turbidity   0
Colour      0
Grade       0
dtype: int64

```

Duplicate Values:

976

Grade Categories:

['high' 'low' 'medium']

In [31... print(df.describe())

```

      pH      Temprature      Taste      Odor      Fat  \
count  1059.000000  1059.000000  1059.000000  1059.000000  1059.000000
mean     6.630123    44.226629    0.546742    0.432483    0.671388
std     1.399679    10.098364    0.498046    0.495655    0.469930
min      3.000000    34.000000    0.000000    0.000000    0.000000
25%      6.500000    38.000000    0.000000    0.000000    0.000000
50%      6.700000    41.000000    1.000000    0.000000    1.000000
75%      6.800000    45.000000    1.000000    1.000000    1.000000
max      9.500000    90.000000    1.000000    1.000000    1.000000

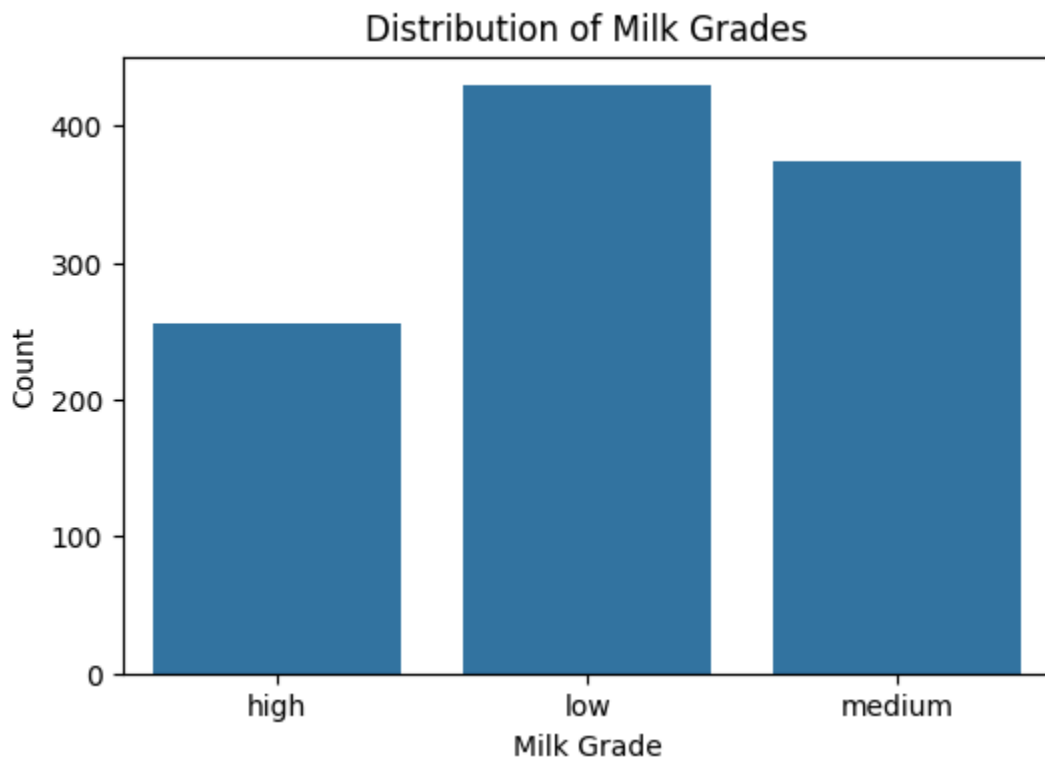
      Turbidity      Colour
count  1059.000000  1059.000000
mean     0.491029   251.840415
std     0.500156    4.307424
min      0.000000   240.000000
25%      0.000000   250.000000
50%      0.000000   255.000000
75%      1.000000   255.000000
max      1.000000   255.000000

```

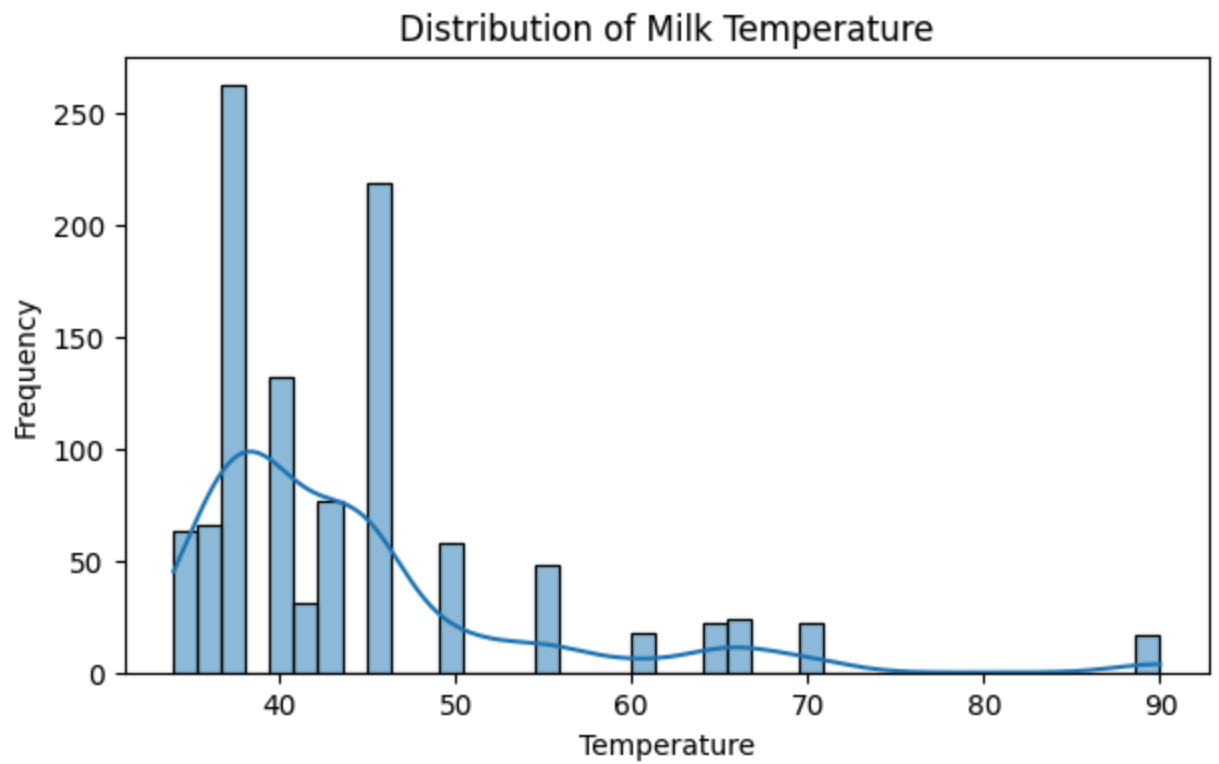
```
In [32...] print(df.groupby("Grade")["Temprature"].describe())
```

	count	mean	std	min	25%	50%	75%	max
Grade								
high	256.0	40.648438	3.739749	35.0	37.0	40.0	45.0	45.0
low	429.0	50.289044	13.043794	34.0	40.0	45.0	55.0	90.0
medium	374.0	39.721925	3.613751	34.0	37.0	38.0	45.0	45.0

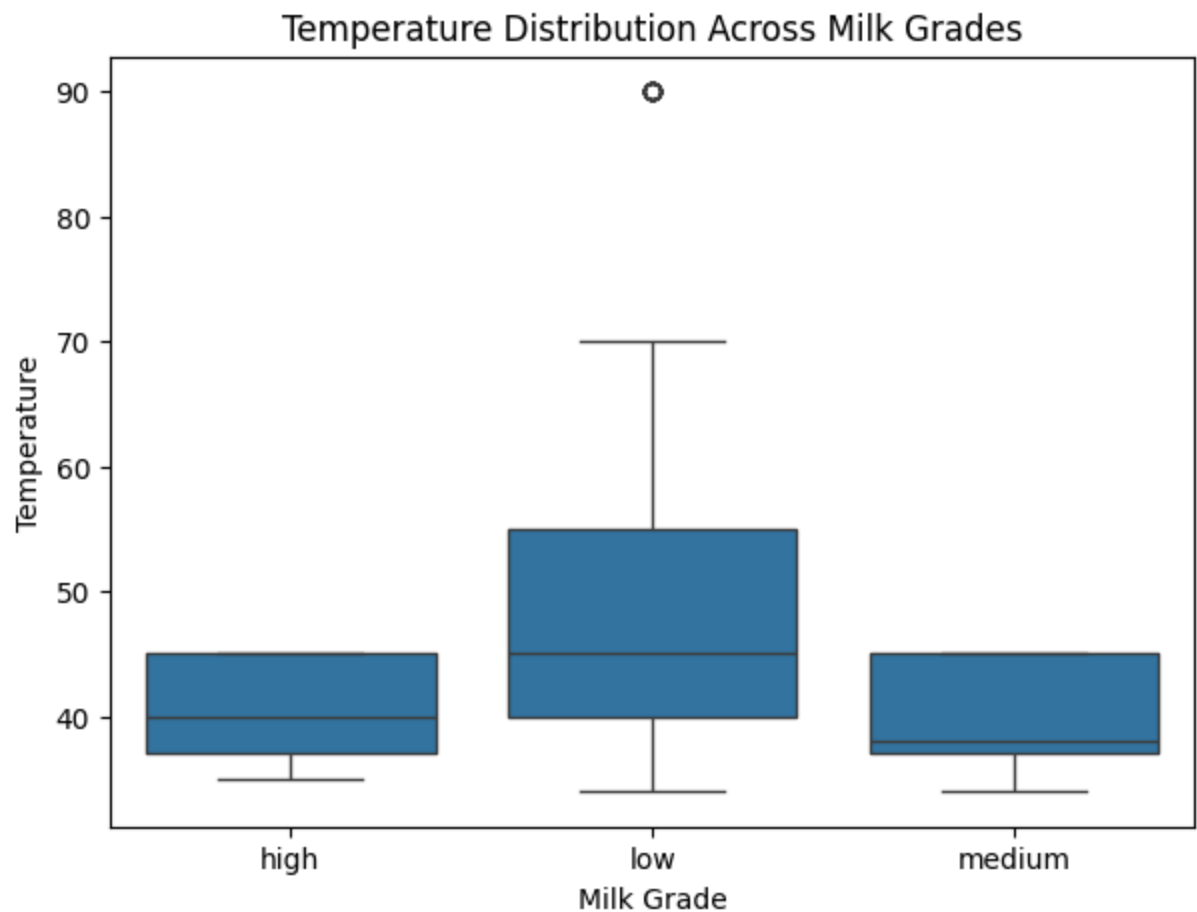
```
In [33...] plt.figure(figsize=(6,4))
sns.countplot(data=df, x="Grade")
plt.title("Distribution of Milk Grades")
plt.xlabel("Milk Grade")
plt.ylabel("Count")
plt.show()
```



```
In [34...] plt.figure(figsize=(7,4))
sns.histplot(df["Temprature"], kde=True)
plt.title("Distribution of Milk Temperature")
plt.xlabel("Temperature")
plt.ylabel("Frequency")
plt.show()
```



```
In [35... plt.figure(figsize=(7,5))
sns.boxplot(data=df, x="Grade", y="Temprature")
plt.title("Temperature Distribution Across Milk Grades")
plt.xlabel("Milk Grade")
plt.ylabel("Temperature")
plt.show()
```

```
In [36... grades = df["Grade"].unique()

groups = [
    df[df["Grade"] == grade]["Temperature"]
    for grade in grades
]

f_stat, p_value = stats.f_oneway(*groups)

print("F-statistic:", f_stat)
print("p-value:", p_value)
```

```
F-statistic: 173.06391181235017
p-value: 9.787957245853683e-66
```

```
In [38... alpha = 0.05
```

```
In [39... if p_value < alpha:
    print("Reject the Null Hypothesis")
else:
    print("Fail to Reject the Null Hypothesis")
```

```
Reject the Null Hypothesis
```

```
In [40... for grade in grades:
    data = df[df["Grade"] == grade]["Temprature"]

    mean = data.mean()
    sem = stats.sem(data)
    ci = stats.t.interval(0.95, len(data) - 1, loc=mean, scale=sem)

    print(f"{grade}:")
    print(f"Mean Temperature = {mean:.2f}")
    print(f"95% Confidence Interval = ({ci[0]:.2f}, {ci[1]:.2f})")
    print()
```

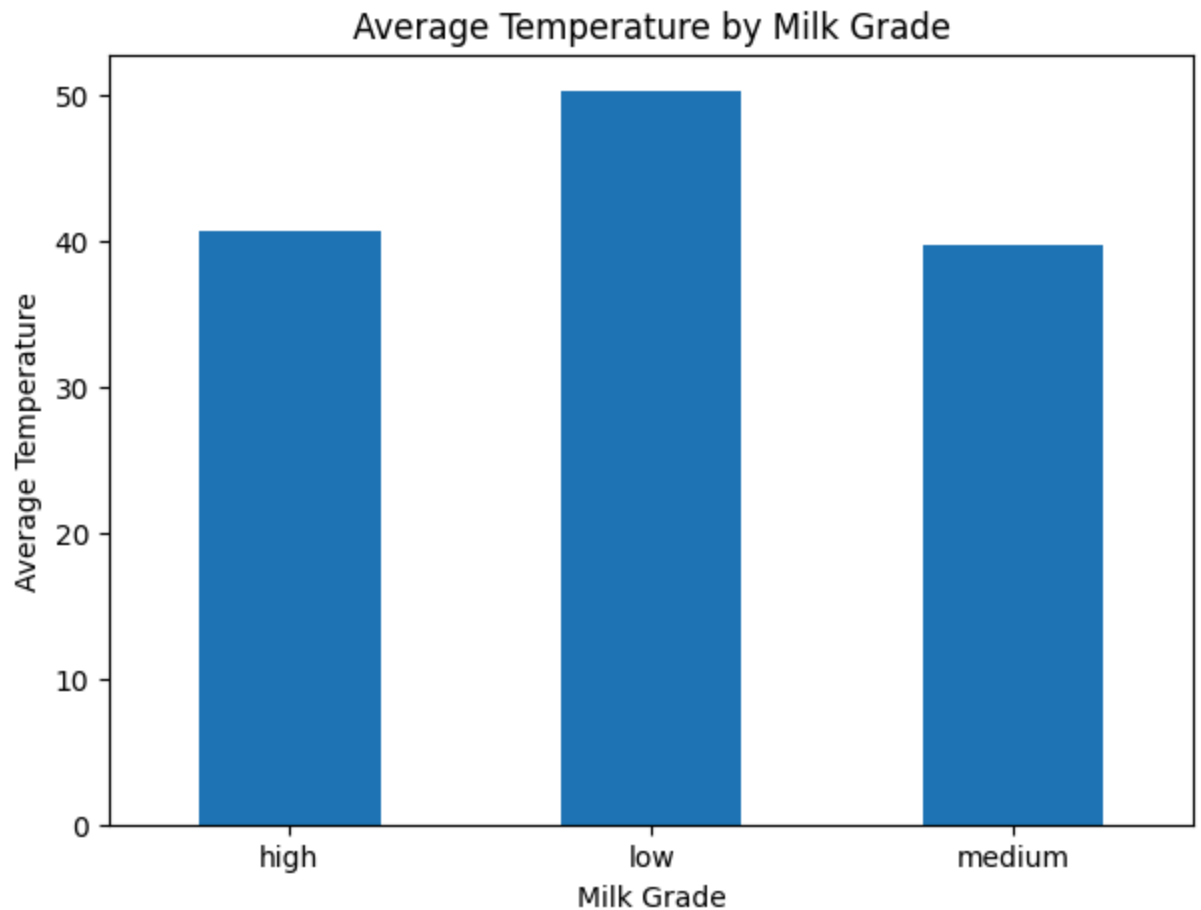
```
high:
Mean Temperature = 40.65
95% Confidence Interval = (40.19, 41.11)
```

```
low:
Mean Temperature = 50.29
95% Confidence Interval = (49.05, 51.53)
```

```
medium:
Mean Temperature = 39.72
95% Confidence Interval = (39.35, 40.09)
```

```
In [41... mean_temp = df.groupby("Grade")["Temprature"].mean()

plt.figure(figsize=(7,5))
mean_temp.plot(kind="bar")
plt.title("Average Temperature by Milk Grade")
plt.xlabel("Milk Grade")
plt.ylabel("Average Temperature")
plt.xticks(rotation=0)
plt.show()
```



```
In [42... print("ANOVA Results")
print("-----")
print("F-statistic:", f_stat)
print("p-value:", p_value)

if p_value < 0.05:
    print("Result: Reject the Null Hypothesis")
    print("There is a statistically significant difference in temperature")
else:
    print("Result: Fail to Reject the Null Hypothesis")
    print("There is no statistically significant difference in temperatur
```

ANOVA Results

F-statistic: 173.06391181235017

p-value: 9.787957245853683e-66

Result: Reject the Null Hypothesis

There is a statistically significant difference in temperature among milk grades.